
Nervous System

Sensory Function

Sensory Physiology

- ~ how the stimulus is transduced by sensory receptors & processed in the brain (examines the **neural consequences of a stimulus**)
- **Sensory Systems** are parts of the NS that receive signals from the environment and the interior of the body and conduct and process these signals
- **Sensory information** is a neural activity originating from stimulation of receptor cells in specific parts of the body

Sensory Physiology

- **Significance of sensory systems**
 1. Exploring the external world:
 - knowledge of the world arises through the senses
 - a. Sensation
 - b. Control of movement
 - c. Maintaining arousal...
 2. Orientation and survival...

Sensory Physiology

- Sensory Modality/Sense

- A group of sensory impressions mediated by a particular organ
 - e.g., ‘Five classic senses’...+...

- Types of sensation/modality

- | | | | |
|---------|-------|-------|-------|
| • Touch | Cold | Pain | Taste |
| • Heat | Light | Sound | Smell |

- Each modality has many constituent qualities or sub-modalities
 - e.g., Taste → sour, salty, sweet, bitter and umami)

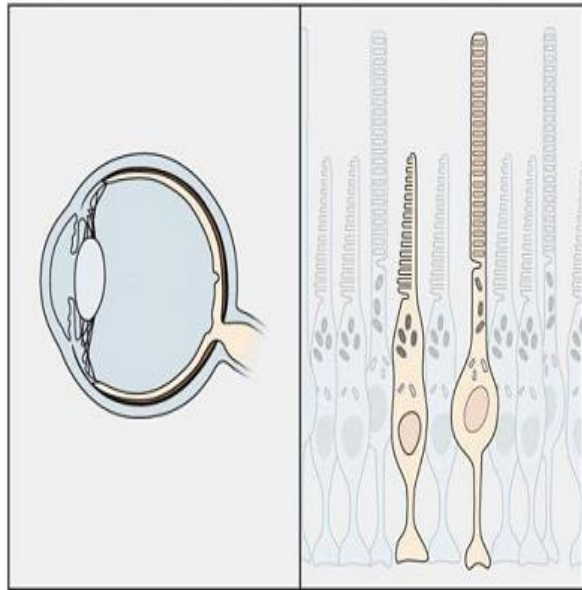
Sensory Physiology

- **Sensory Modalities**
 - Classic five senses
 - Somatic sensations of proprioception + (posture & movement of our own body)
 - Pain + Itch + Temperature
 - Visceral sensations (conscious + subconscious)
 - Vestibular sense of balance (position of body in gravitational field + head movement)
 - ...

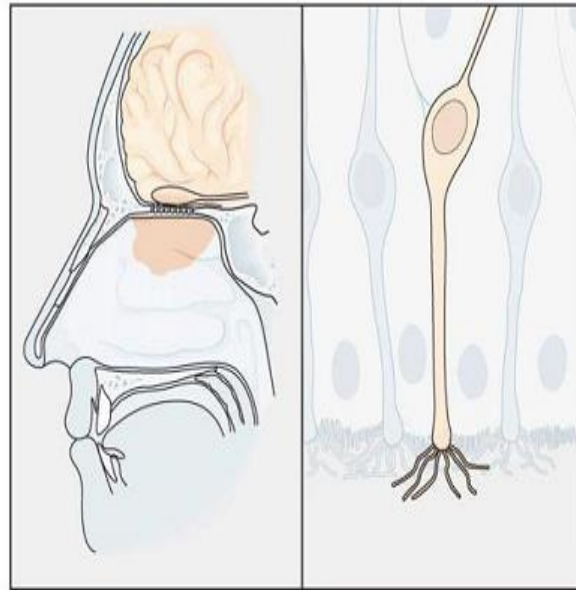
Sensory Physiology

- The major sensory modalities in humans are mediated by distinct classes of receptor neurons located in specific sense organs...

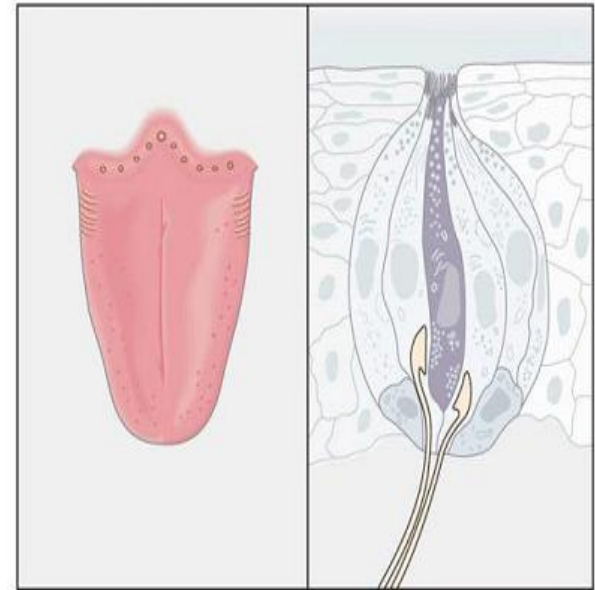
Vision



Smell

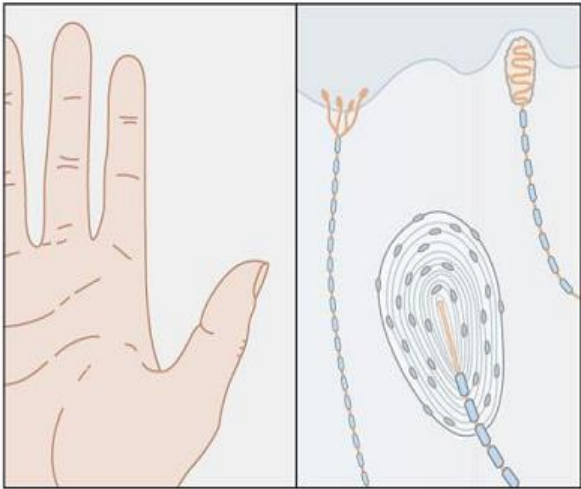


Taste

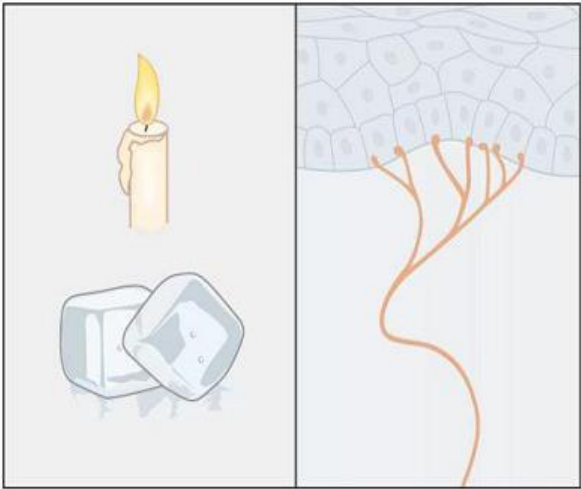


Sensory Physiology

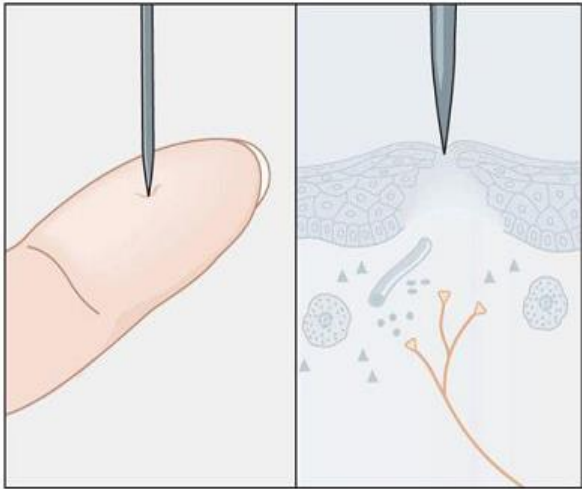
Touch



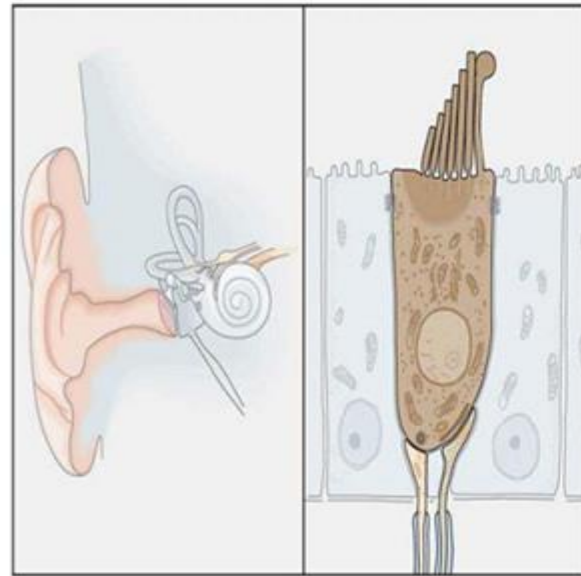
Thermal senses



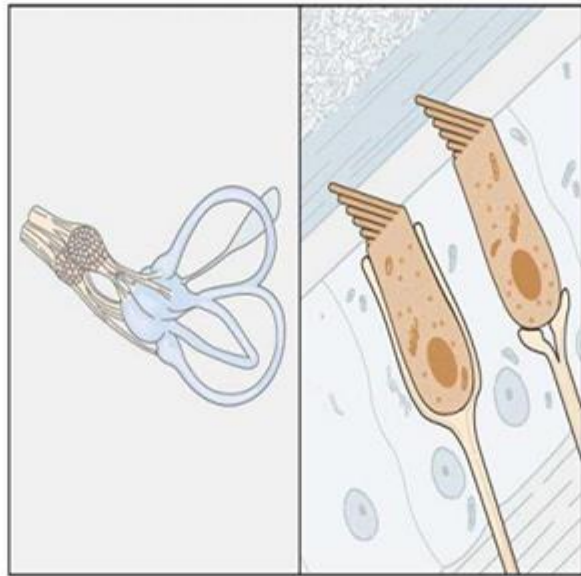
Pain



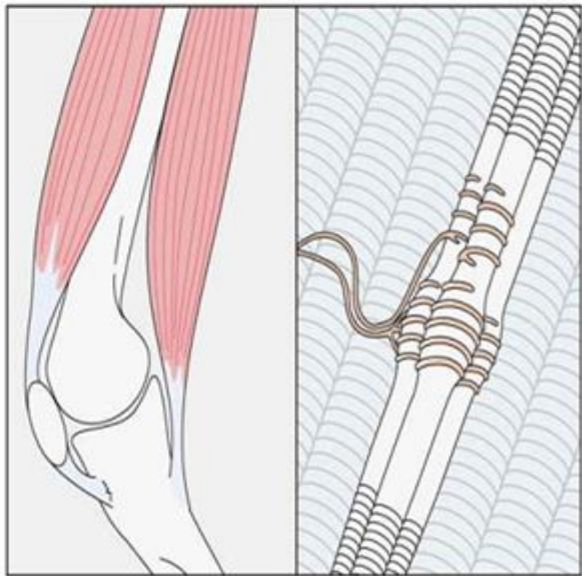
Hearing



Balance

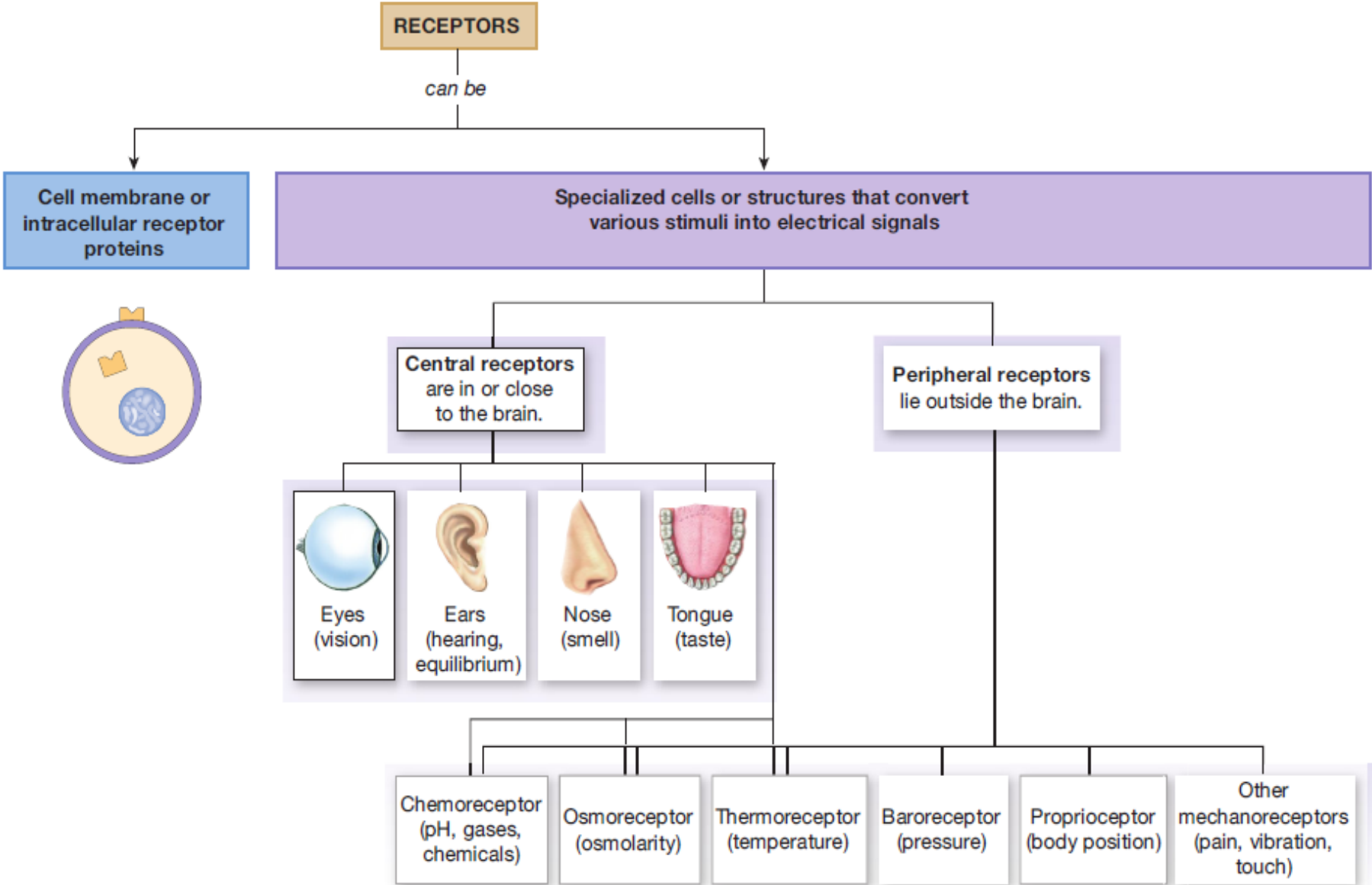


Proprioception



Sensory Physiology

- Receptor
 - **A Transducer** (a neuron or a specialized cell) that converts a particular form of energy in the environment into electrical signals in neurons
 - Located at sites exposed to their specific stimuli
 - *Receptor specificity* → Specialization of receptors for specific qualities (**not absolute**)



Classification of Sensory Receptors

Modality	Stimulus	Receptor types	Receptors
Vision	Light (photons)	Photoreceptor	Rods & Cones
Audition	Sound (pressure waves)	Mechanoreceptor	Hair cells (cochlear)
Balance	Head motion, gravity and acceleration	Mechanoreceptor	Hair cells (semicircular canals)
Taste	Chemical	Chemoreceptor	Taste buds
Smell	Chemical	Chemoreceptor	Olfactory sensory n
Touch	Skin deformation and motion	Mechanoreceptor	Skin receptors
Proprioception	Muscle length and force, Joint angle	Mechanoreceptor	Muscle spindles and Joint capsules

Sensory Physiology

- Stimulus
 - A factor that elicits a sensory impression of a certain quality
 - Measurable quantity
 - A stimulus acquires its quality by virtue of the reaction with the stimulus detecting cells of the sense organs (*the receptors*)
- Environmental signals → mechanical force, light, sound, chemicals, and temperature...

Sensory Physiology

- Origin/Source of stimulus

1. Exteroceptors

- Receive stimuli from outside of the body
- Ear, eye, taste...

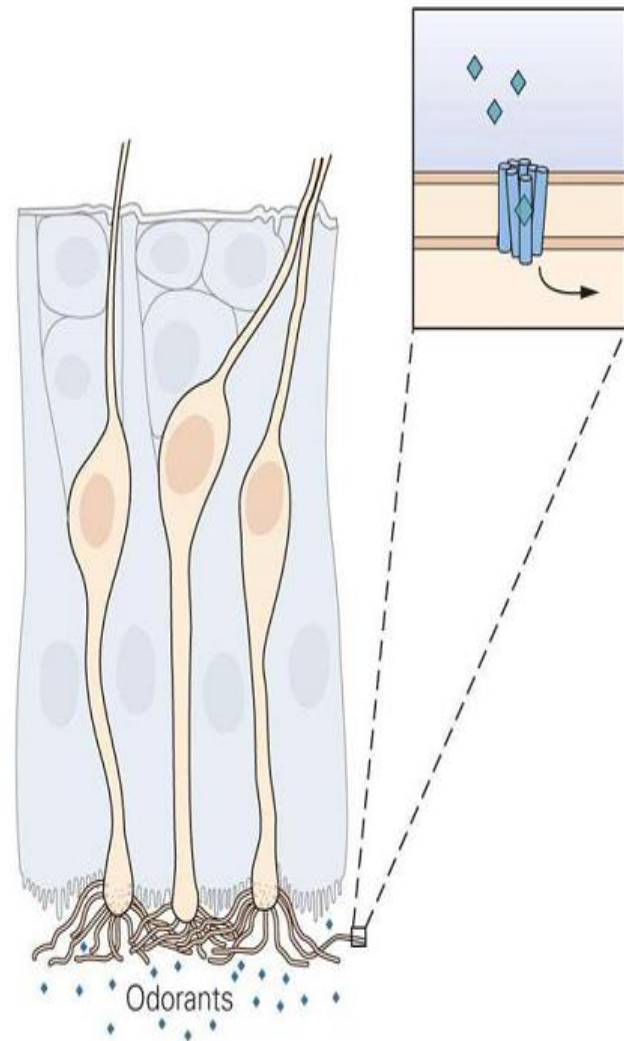
2. Enteroceptors

- Receive stimuli within the body
- Chemoreceptors → blood gases (O₂, CO₂, pH, osmolality)
- Baroreceptors → blood pressure
- Proprioceptors → position of the limbs (muscle contraction)

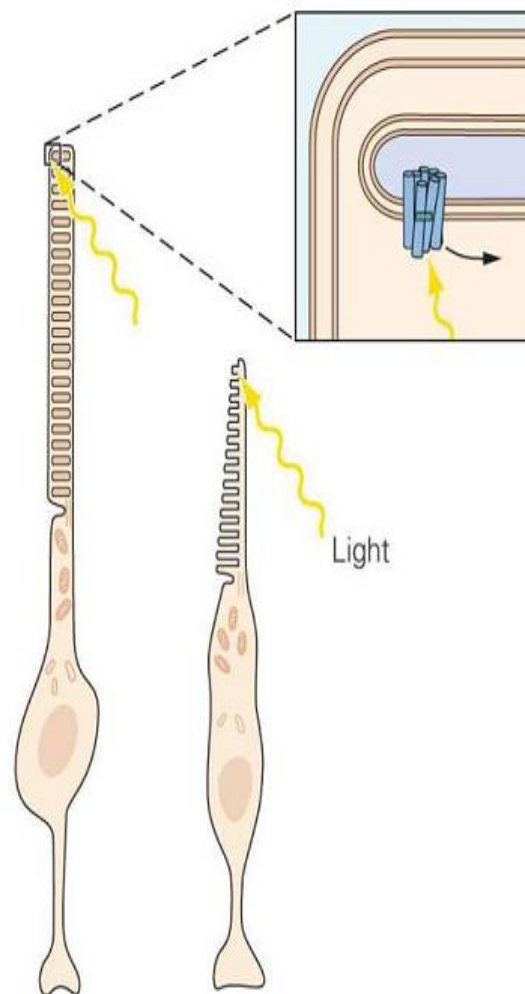
Sensory Physiology

- Type of stimulus energy
 1. Mechanoreceptors → detect deformation (pressure) and sounds
 - Pacinian corpuscles
 - Stretch receptors
 - Hair cells...
 2. Thermoreceptors → detect environmental temperature (+ *hypothalamus...*)
 3. Nociceptors → detect tissue damage/ injury (**Pain**)

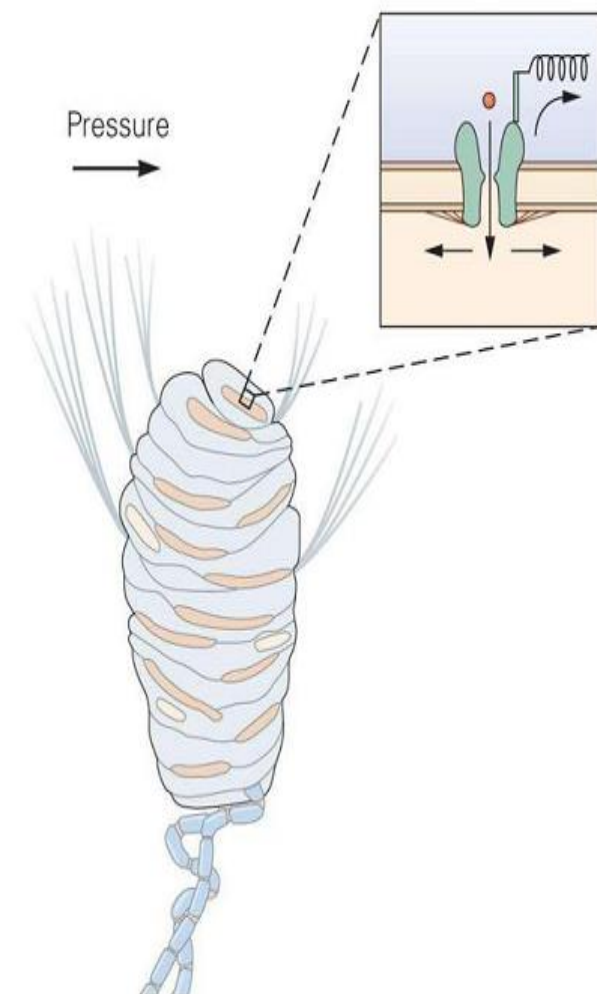
A Chemoreceptor



B Photoreceptor



C Mechanoreceptor

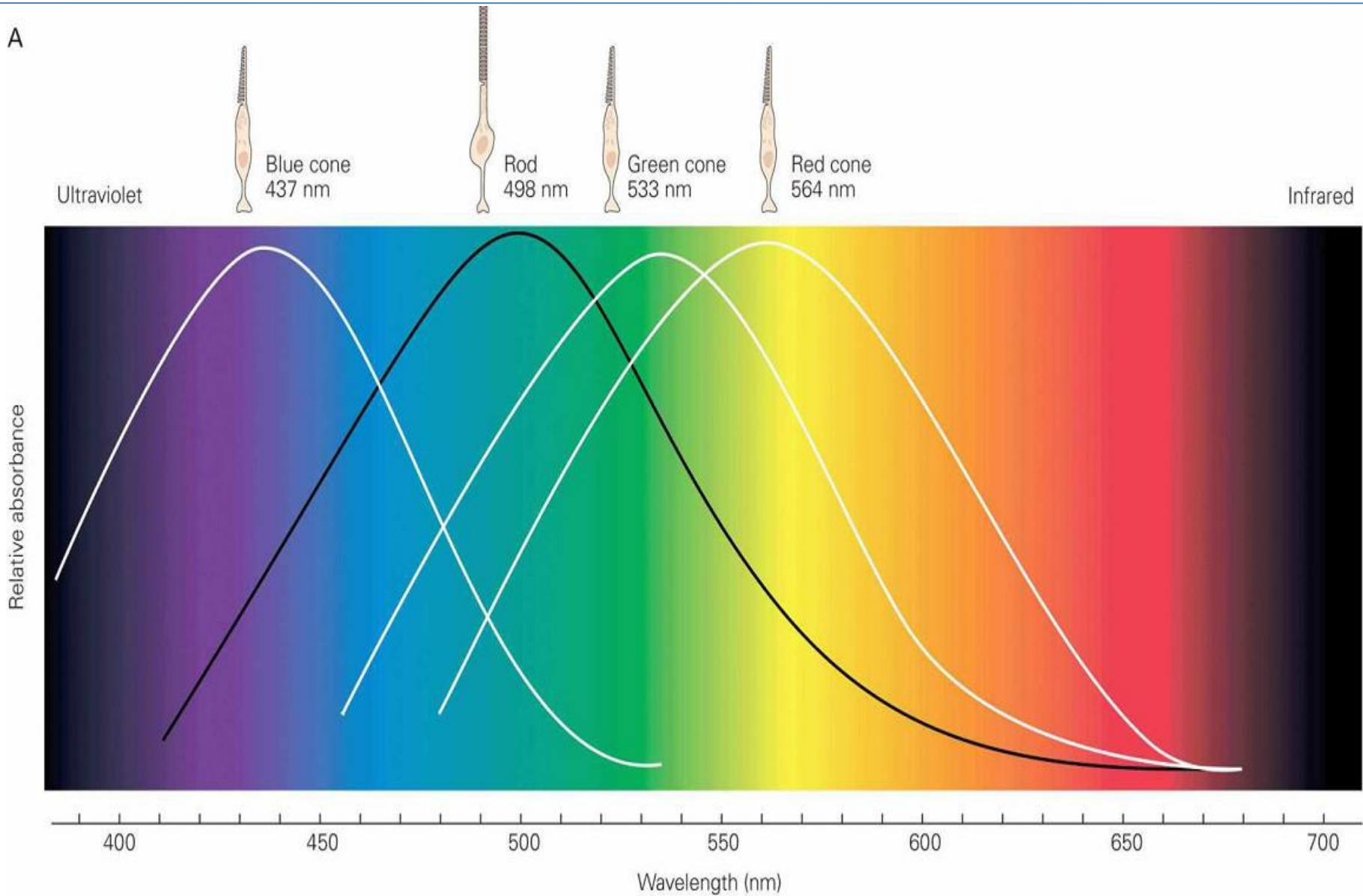


Sensory receptors are specialized to transduce a particular type of stimulus energy into electrical signals.

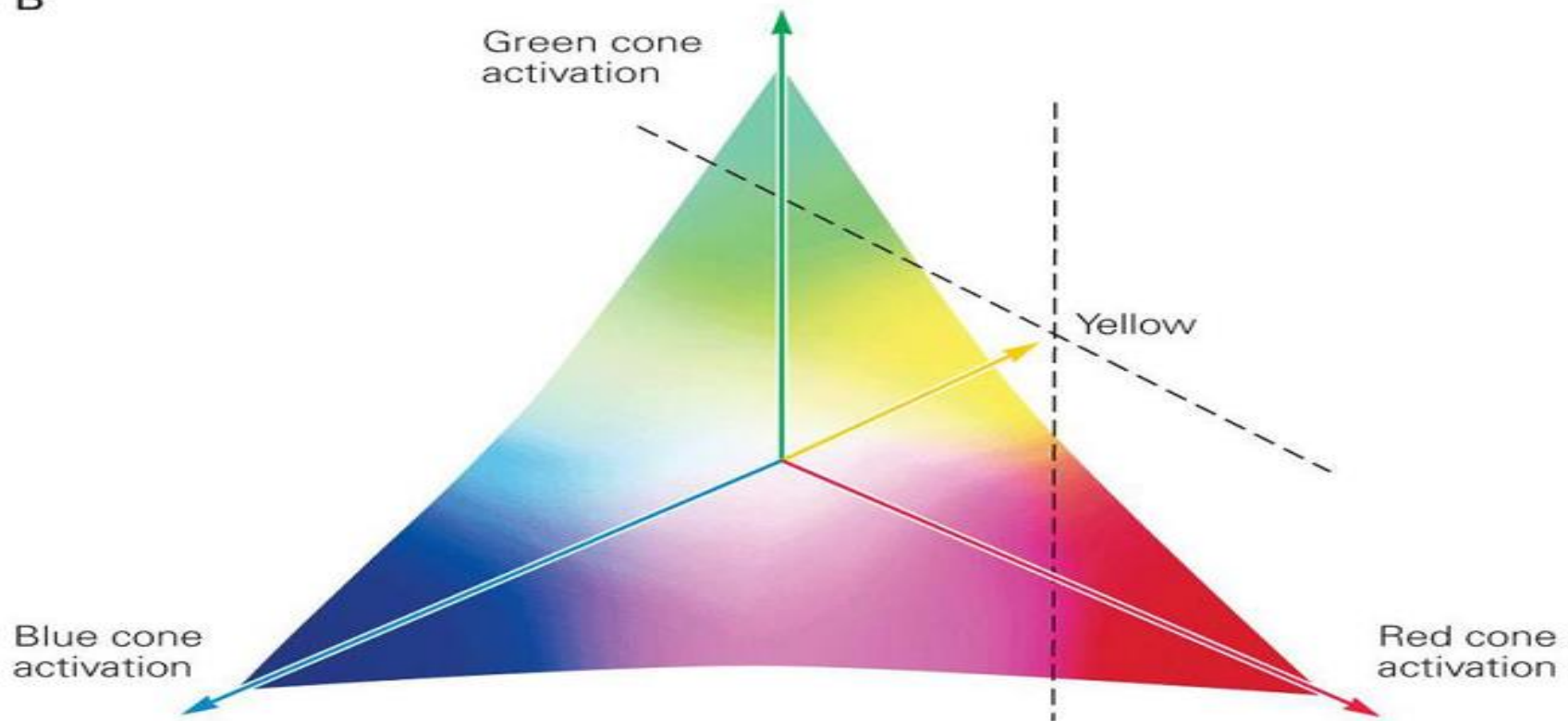
Sensory Physiology

- Type of stimulus energy
 4. Chemoreceptors → detect the type/level of chemicals
 - Olfactory receptors
 - Gustatory receptors
 - Osmoreceptors
 - Carotid body receptors
 5. Photoreceptors → detect light (390-670nm)
 - Rods and Cones (*blue, green, red*)

A



B



- Human perception of colors results from the simultaneous activation of three different classes of photoreceptors in the retina...
- We are able to perceive a spectrum of colors through varying combinations of photoreceptors

Sensory Physiology

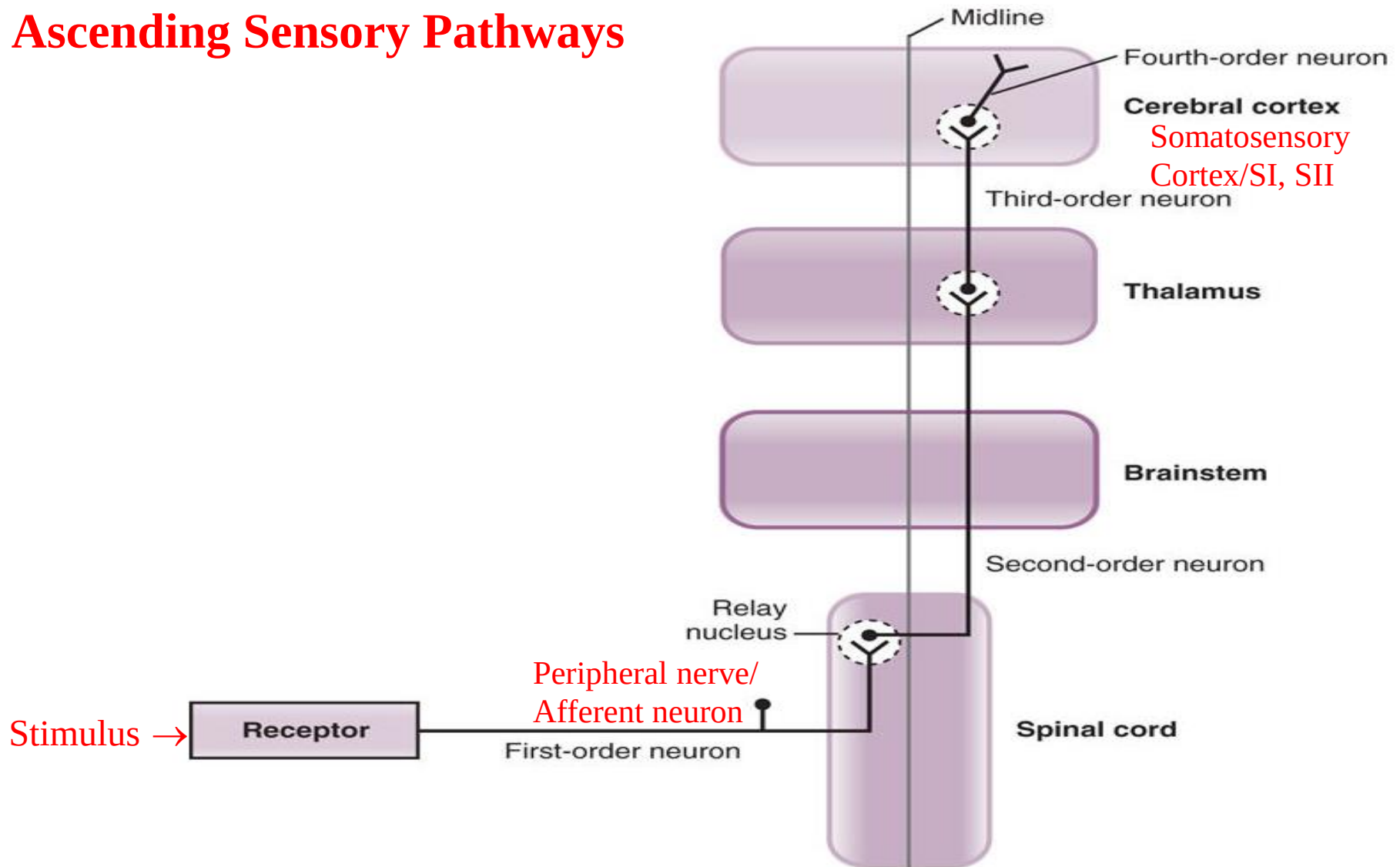
- Functions of a receptor

1. *Reception* → ability of a receptor to absorb energy of a stimulus
 - Sensations and perceptions begin with sensory reception, detection of stimuli by sensory receptors
2. *Transduction* → conversion of stimulus energy into E_m
 - Sensation involves converting stimulus energy into change in E_m of sensory receptors
3. *Integration* → processing of receptor potentials
4. *Transmission* → receptor potentials transmitted via APs to CNS

Ascending Sensory Pathways

- Any sensory system going to the cerebral cortex cross over (decussate) at some point (spinal cord or brain stem/medulla)
- Ascending sensory pathways
 1. Dorsal column-medial lemniscus pathway
 2. Spinothalamic pathway

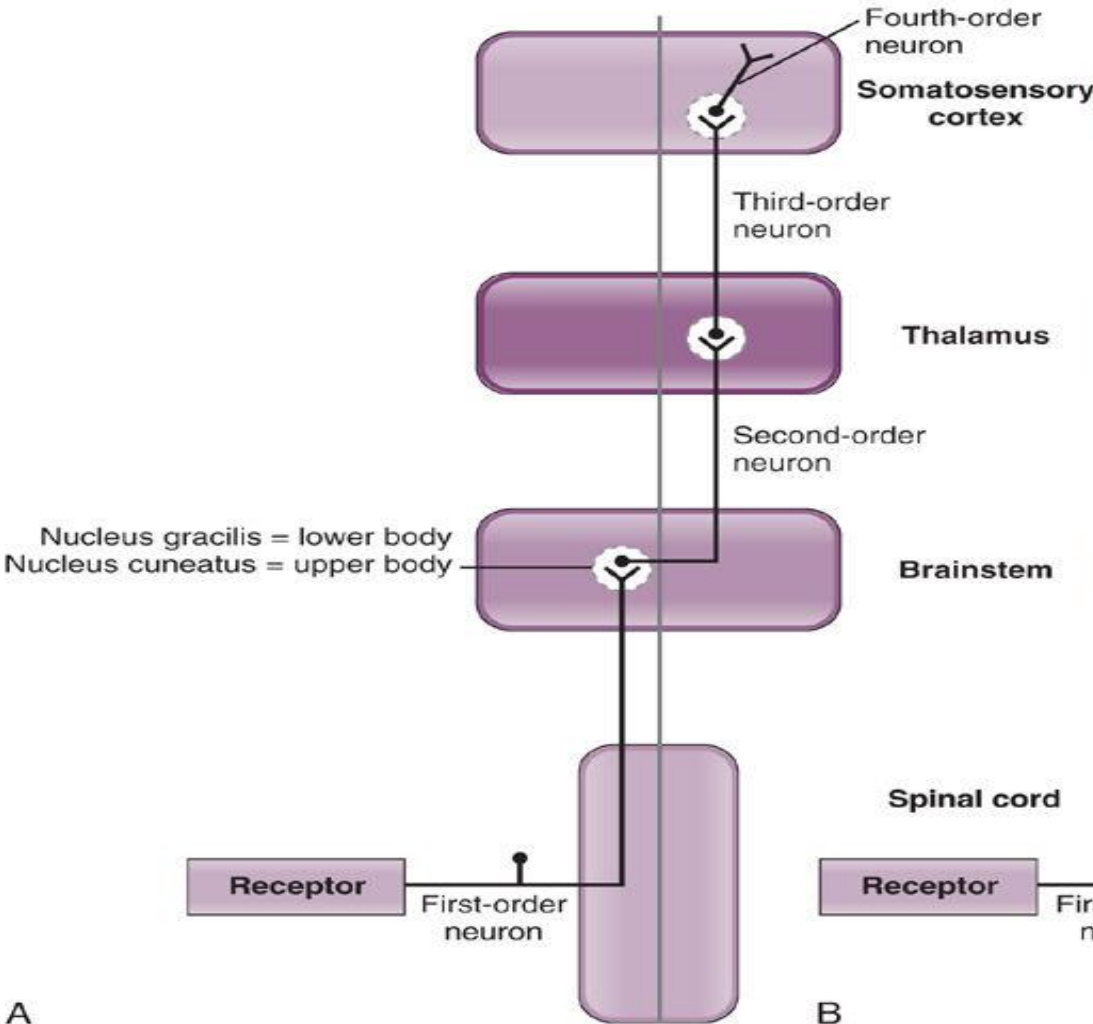
Ascending Sensory Pathways



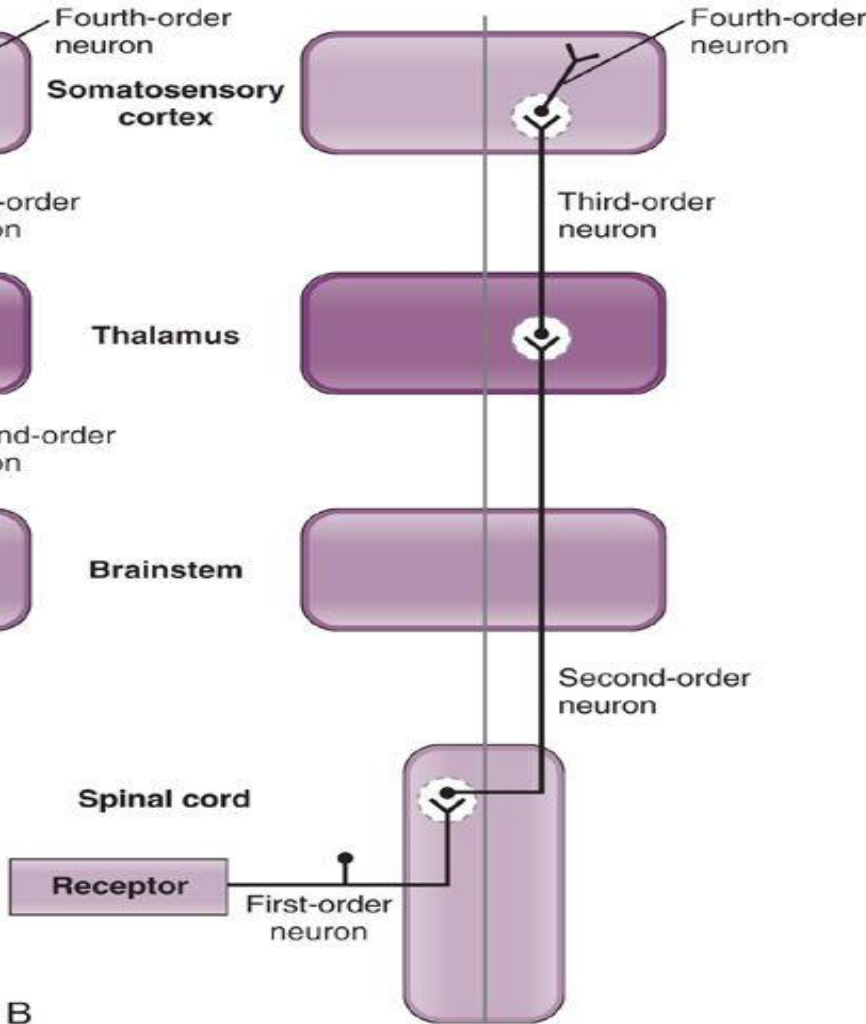
Sensory pathways from the sensory receptors to the cerebral cortex

Two Ascending Sensory Pathways

1 Dorsal column system
(fine touch, pressure, proprioception)



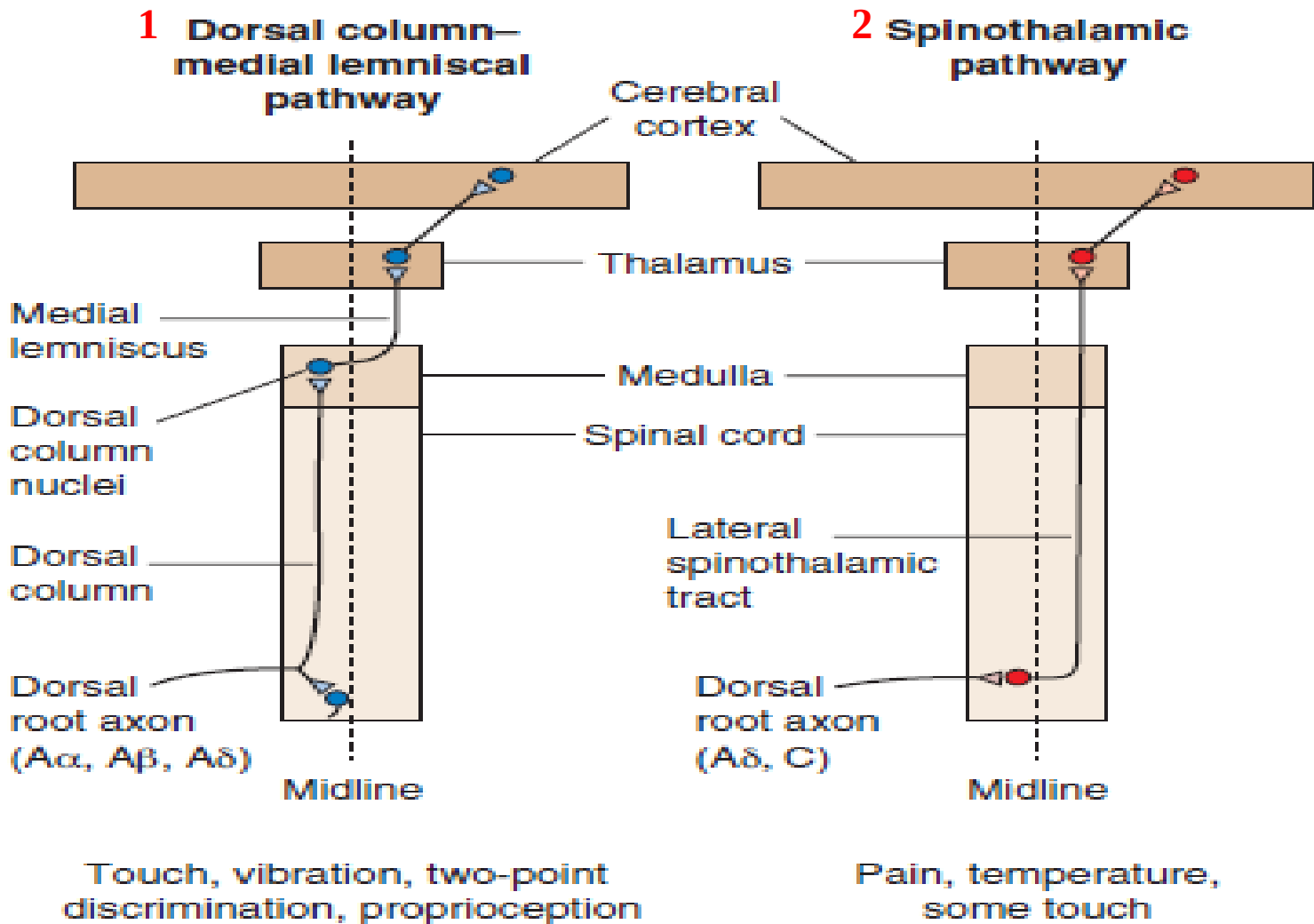
2 Spinothalamic system
(pain, temperature, crude touch)



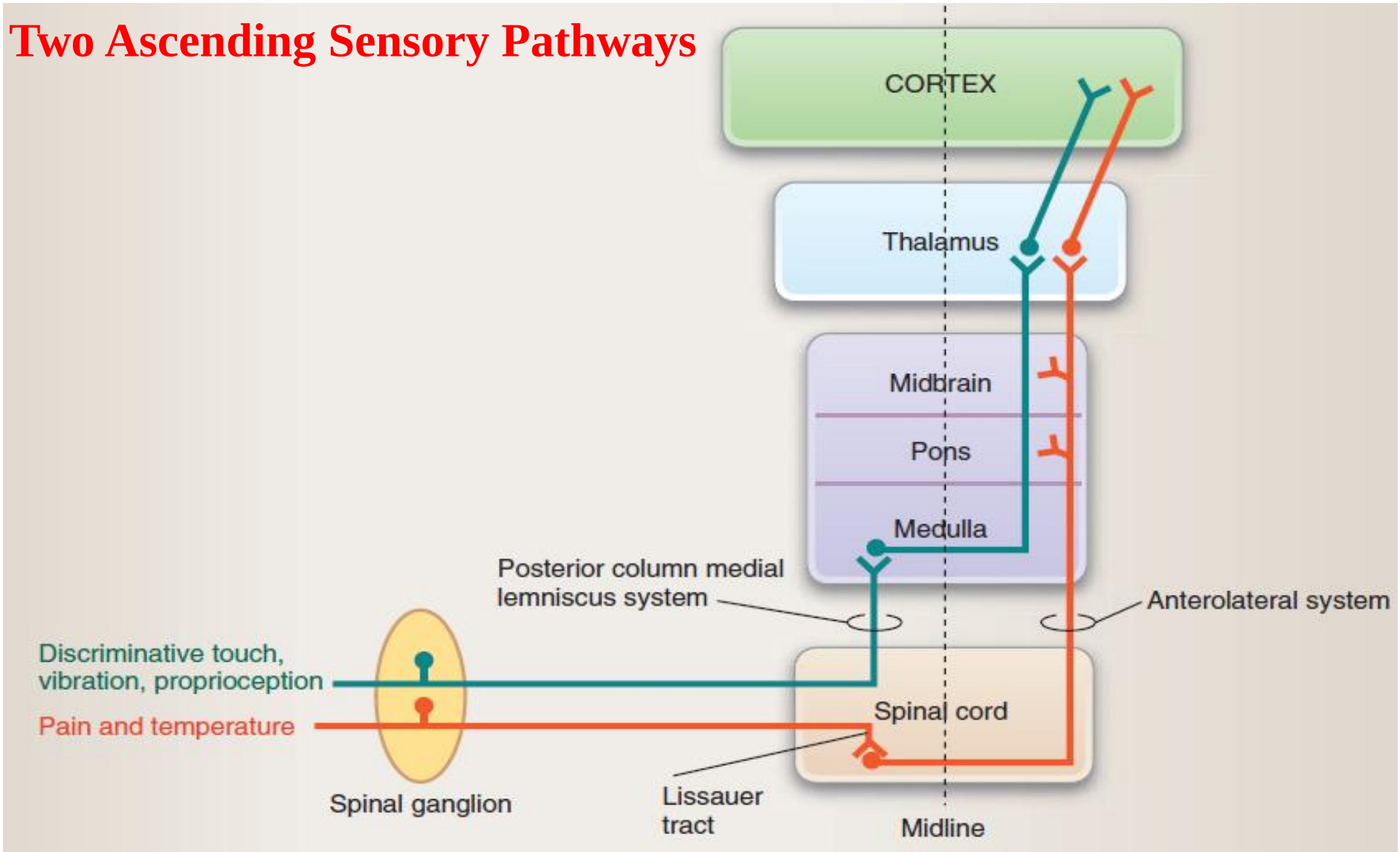
A

B

Two Ascending Sensory Pathways

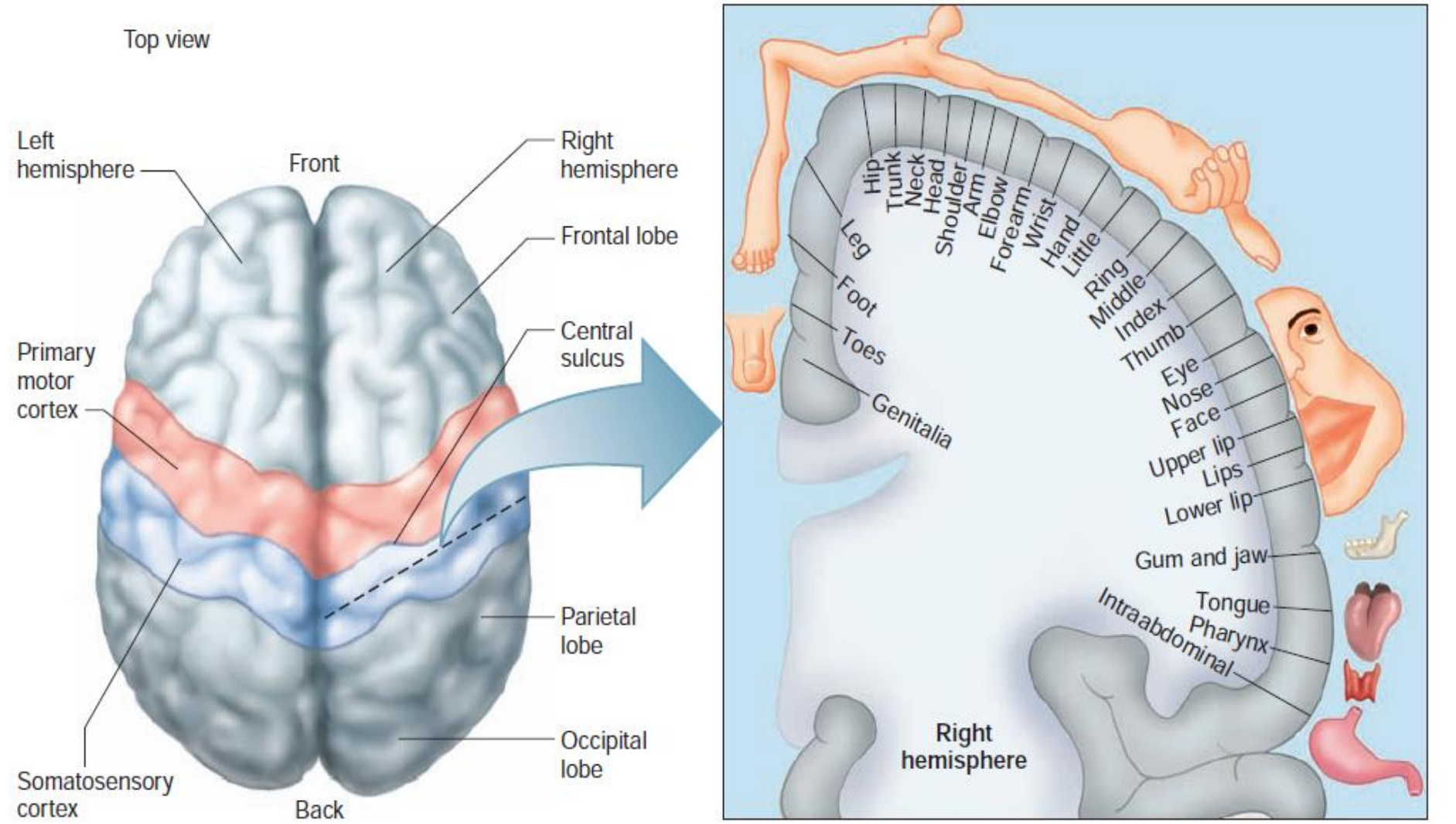


Two Ascending Sensory Pathways

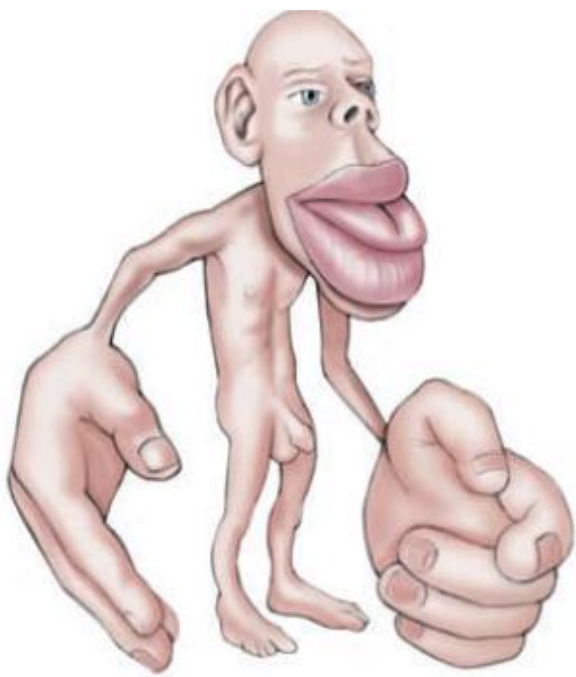
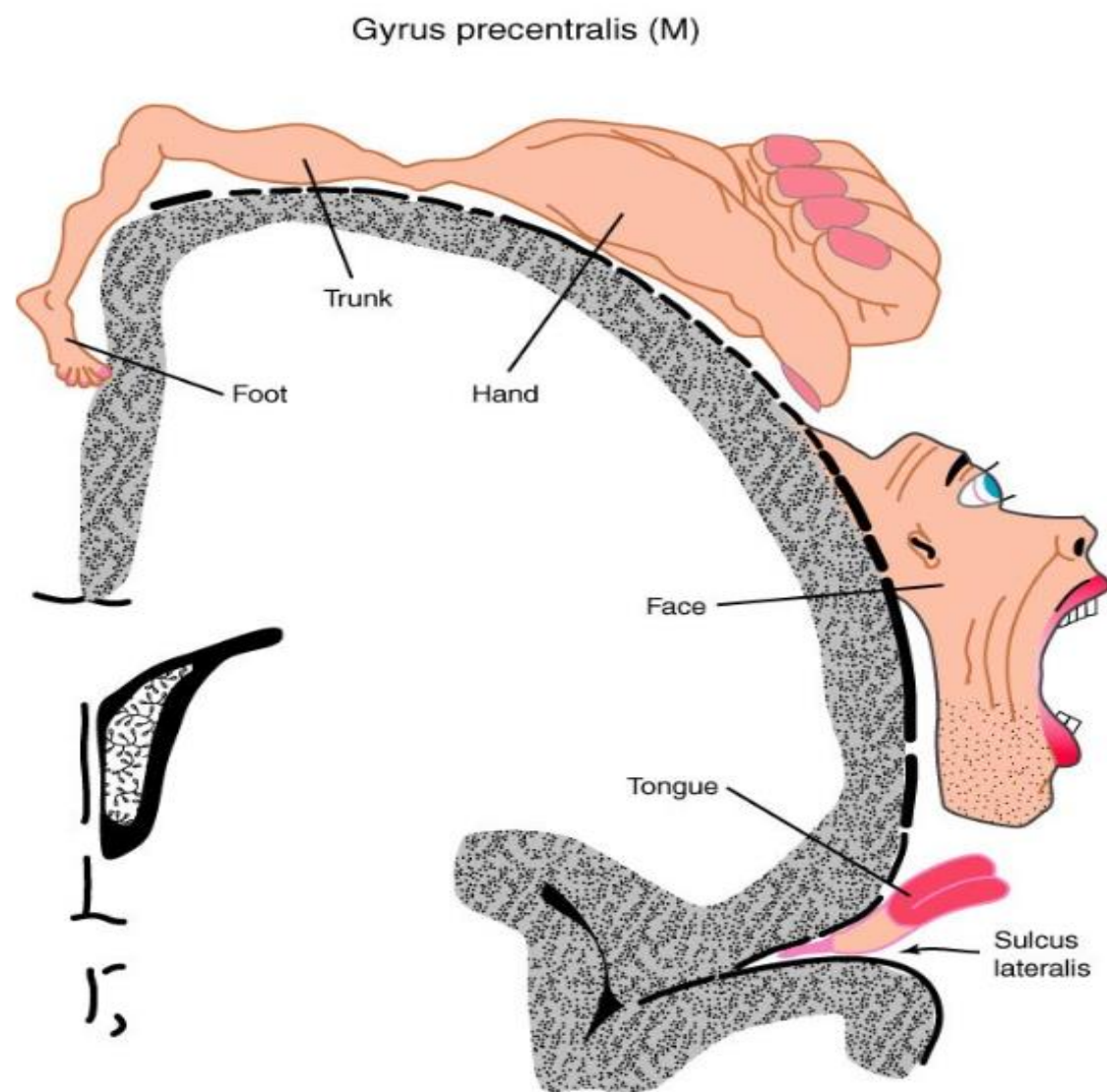


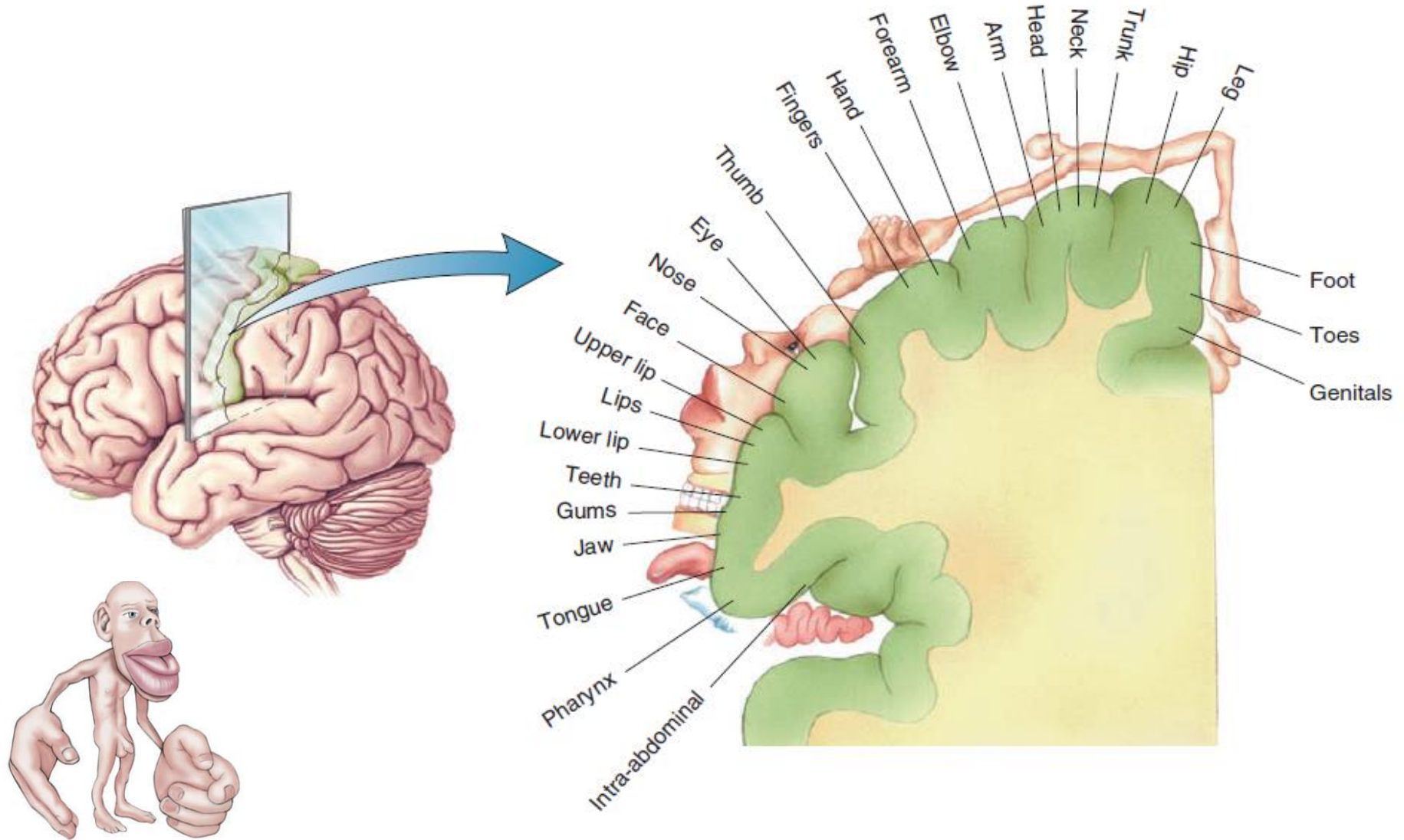
Conscious sensory input and tracts to the cortex

Somatosensory Homunculus

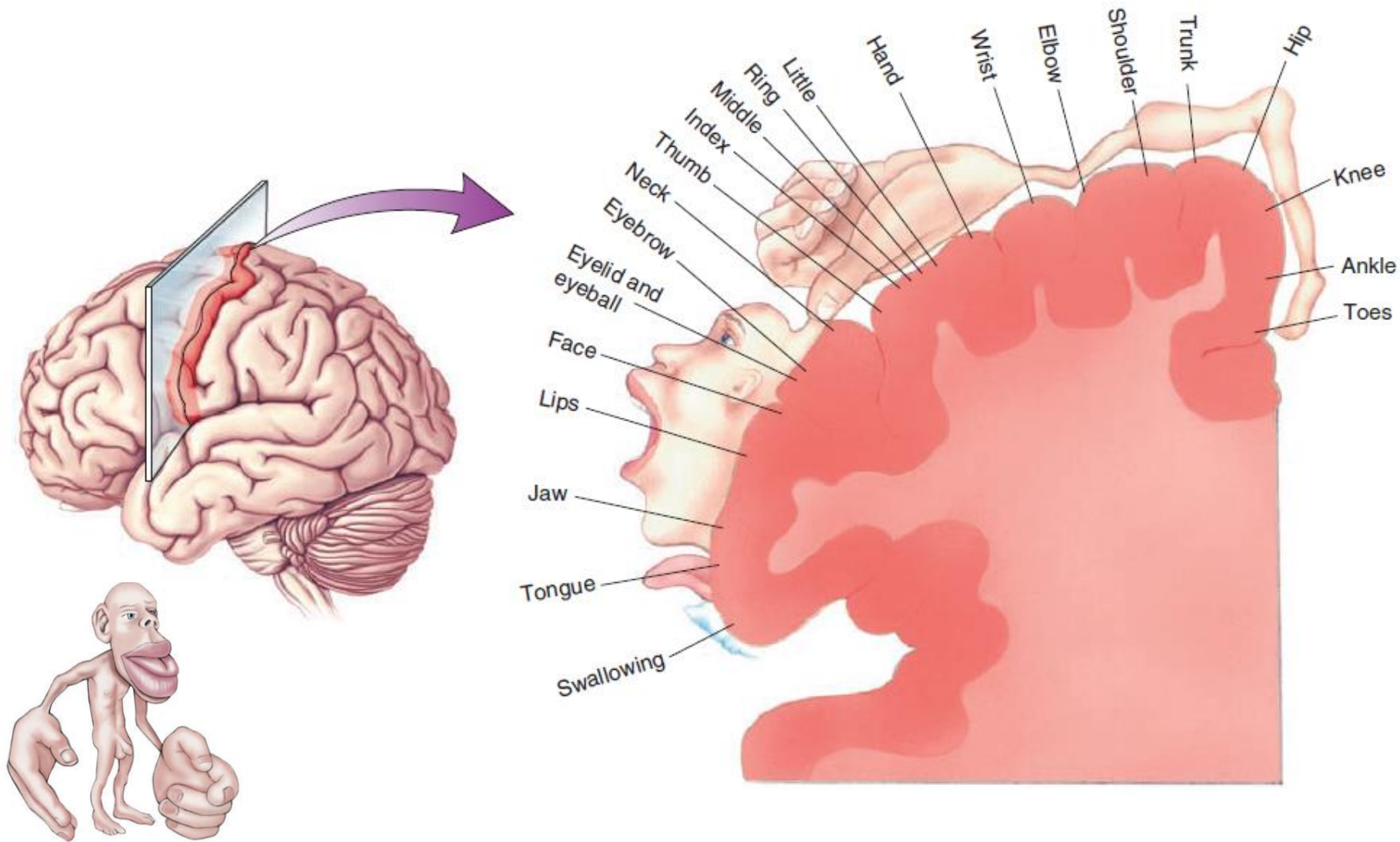


Somatosensory Homunculus





Somatotopic map of the body surface onto primary somatosensory cortex



Somatotopic map of the human primary motor cortex